Study B — one-page summary: do the three methods differ in community composition?

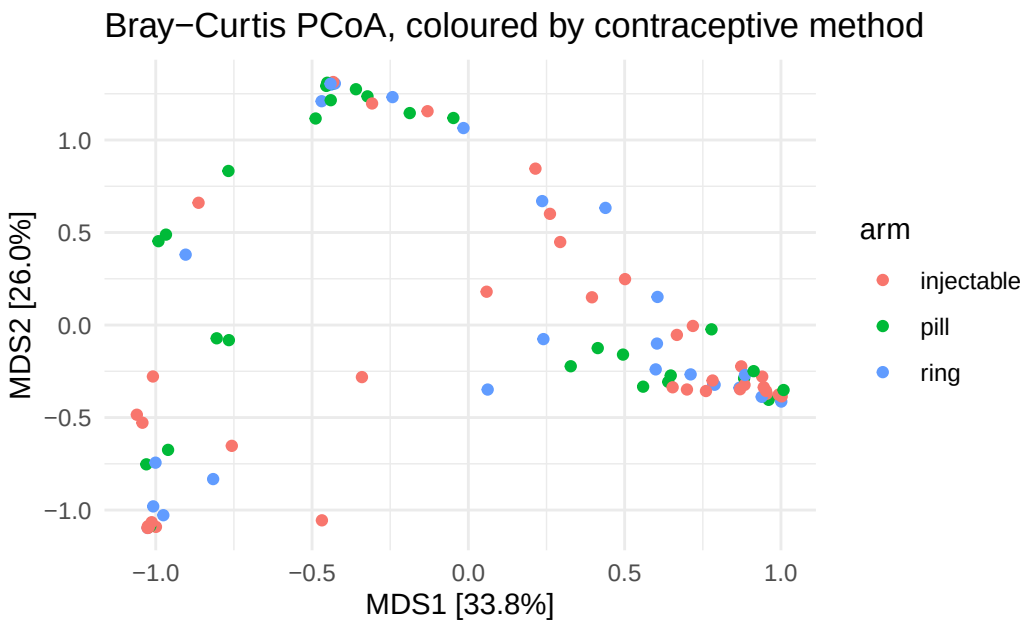
The full workings live in `02-community-composition.qmd` (ordination and PERMANOVA) and `03-bv-by-arm.qmd` (the BV check). This page is the version to present: the answer, the two figures that carry it, and the numbers behind it.

The answer

No — contraceptive method explains almost none of the variation in these vaginal communities (~3%, p 0.15). What does explain it is BV status (~14%, p = 0.001), and BV is spread evenly across the arms. The one suggestive exception is pill vs injectable (p 0.03, but 0.10 once the three comparisons are corrected for), which agrees with a simple biology check and is tomorrow's lead.

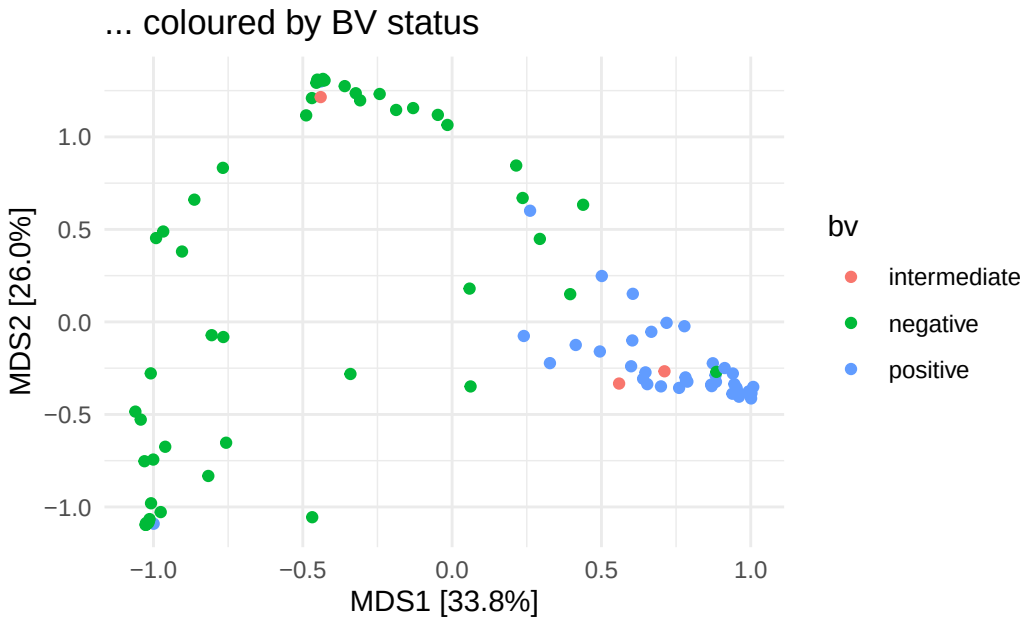
Figure 1 — the map: three overlapping crowds, not three islands

Each point is one participant's vaginal community. Bray-Curtis PCoA: points close together have similar communities.



93 samples & 220 taxa (unique). PCoA tax_transform=compositional dist=bray

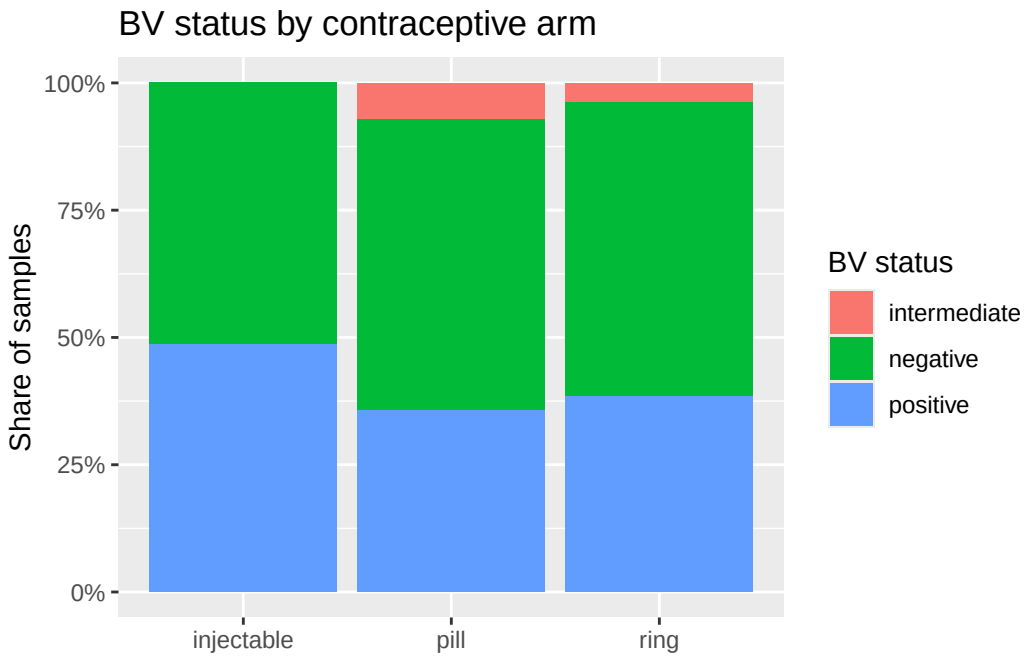
The same points, coloured by what actually separates them:



93 samples & 220 taxa (unique). PCoA tax_transform=compositional dist=bray

Figure 2 — BV is everywhere, in every arm

BV prevalence by arm; Fisher's exact test p 0.45 (details in `03-bv-by-arm.qmd`). No arm is protected or penalised.



The evidence, in one table

Claim	Number	Where it comes from
Arm alone	$R^2 = 0.033$, p = 0.147	PERMANOVA, `02-community-composition`
Arm after BV, pH, STI and batch are accounted for	$R^2 = 0.027$, p = 0.108	PERMANOVA, `02-community-composition`
BV status's share of the variation between samples	$R^2 = 0.143$, p = 0.001	PERMANOVA, `02-community-composition`
Pill vs injectable, tested alone	$R^2 = 0.039$, p = 0.033	Pairwise PERMANOVA, `02-community-composition`
Same overall test, 9 low-depth samples dropped	$R^2 = 0.057$, p = 0.011	Sensitivity analysis, `02-community-composition`
Lactobacillus share, pill vs other arms (median)	84.8% vs 45.0%	Biology check, `02-community-composition`

The one-liner for the slides

> Overall, no: method explains ~3% of the variation in community composition (p 0.15); BV explains ~14% and is evenly spread across arms (p 0.45). Pill vs injectable is the suggestive exception (p 0.03, 0.10 after Holm correction; Lactobacillus share 85% vs ~45%) — chase it with the gene table.

Choices made

- All methods, thresholds and seeds are inherited from the two full reports (`02-community-composition.qmd`, `03-bv-by-arm.qmd`); nothing was re-decided here — this page only re-presents them.
- Main distance **Bray-Curtis on relative abundances** (compositional transform), matching `02-community-composition.qmd`; Aitchison and Jaccard agreed in the full report and are not repeated here.
- **999 permutations, seed 202** for every PERMANOVA on this page.
- Figures shown are **arm-coloured and BV-coloured PCoA** plus the BV bar chart; alternatives (pH, STI, batch panels) are in the full report and changed nothing.
- The evidence table **recomputes** the numbers with the same seeds rather than copy-pasting them, so it would visibly disagree if the underlying reports were changed.